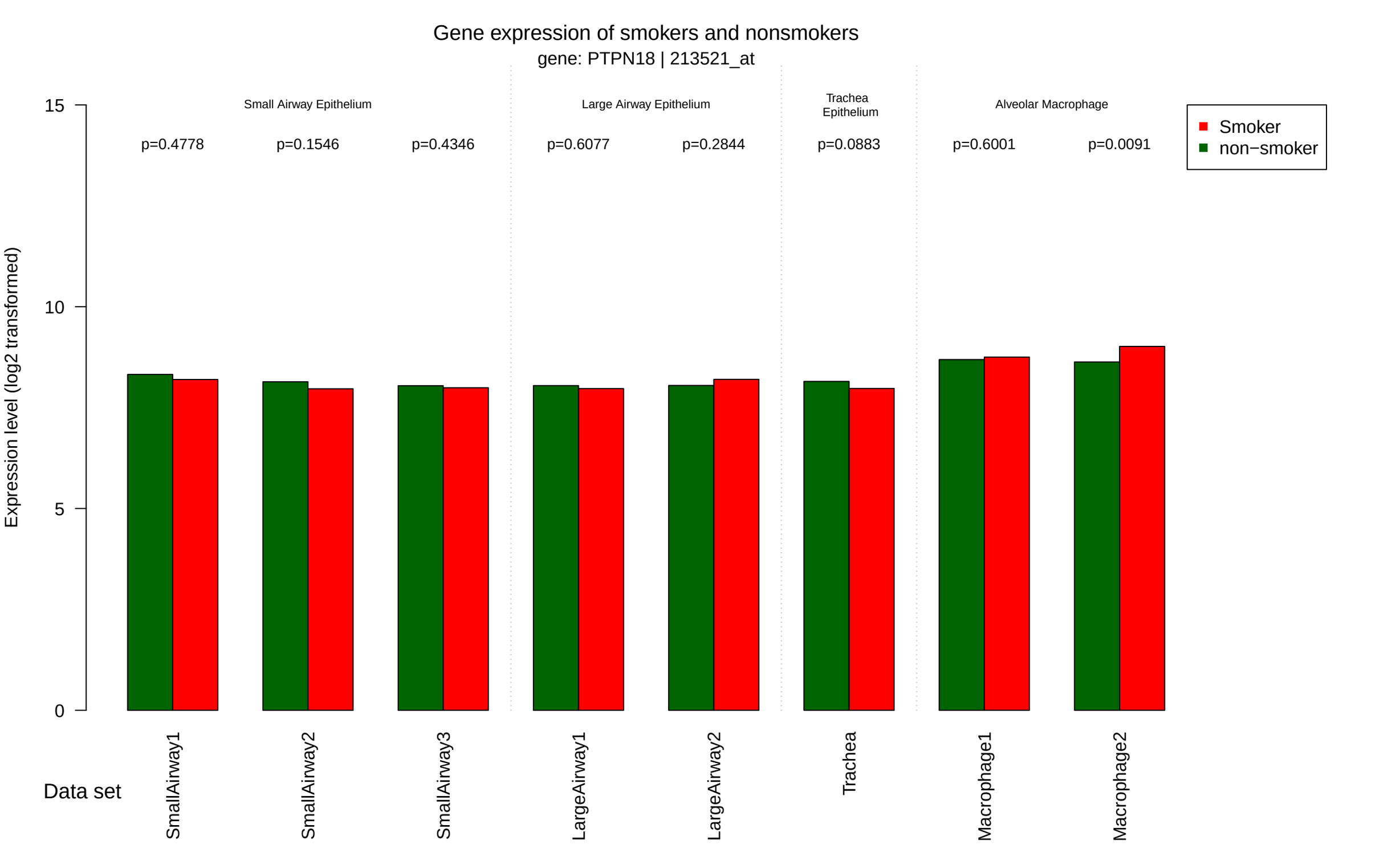
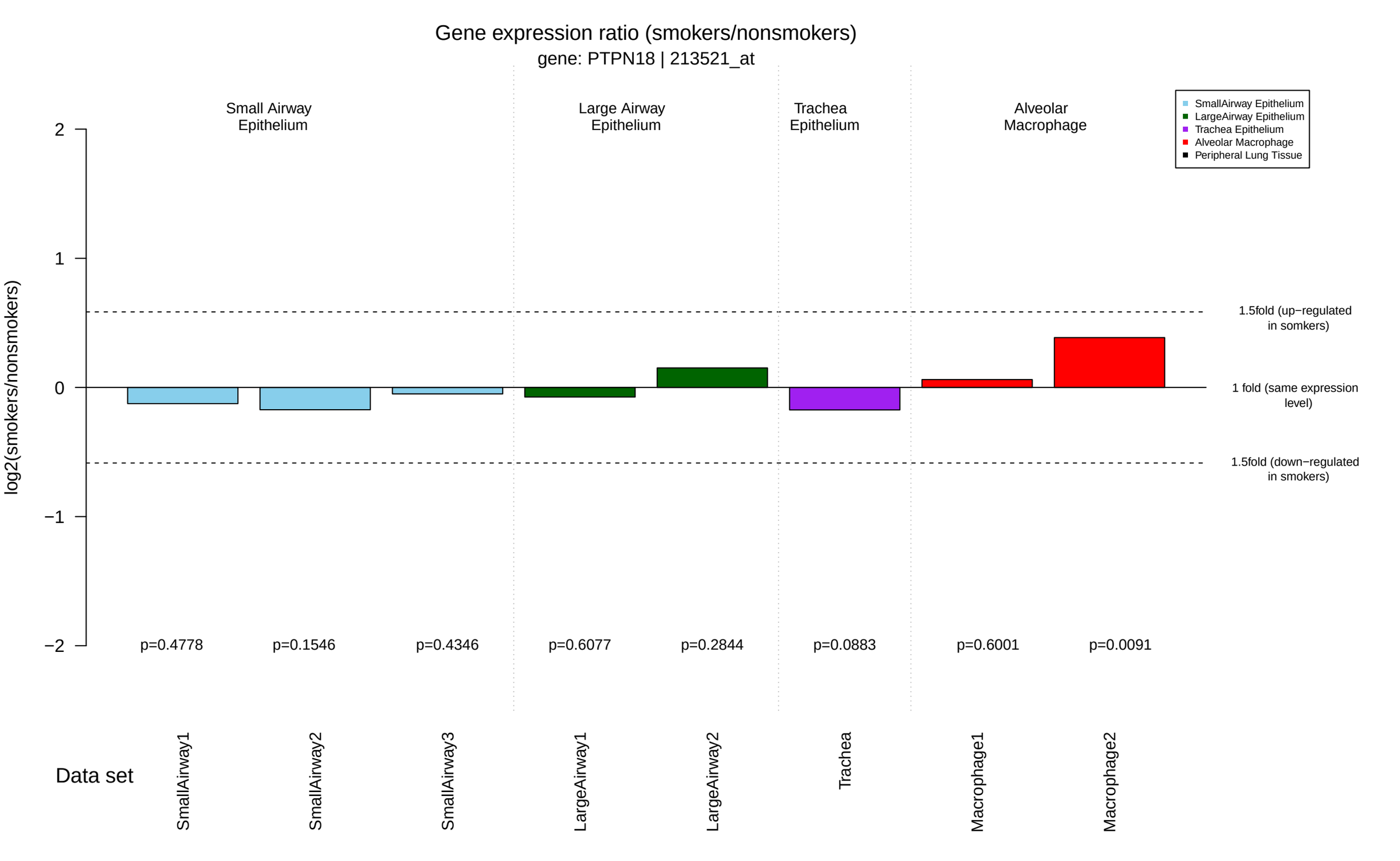
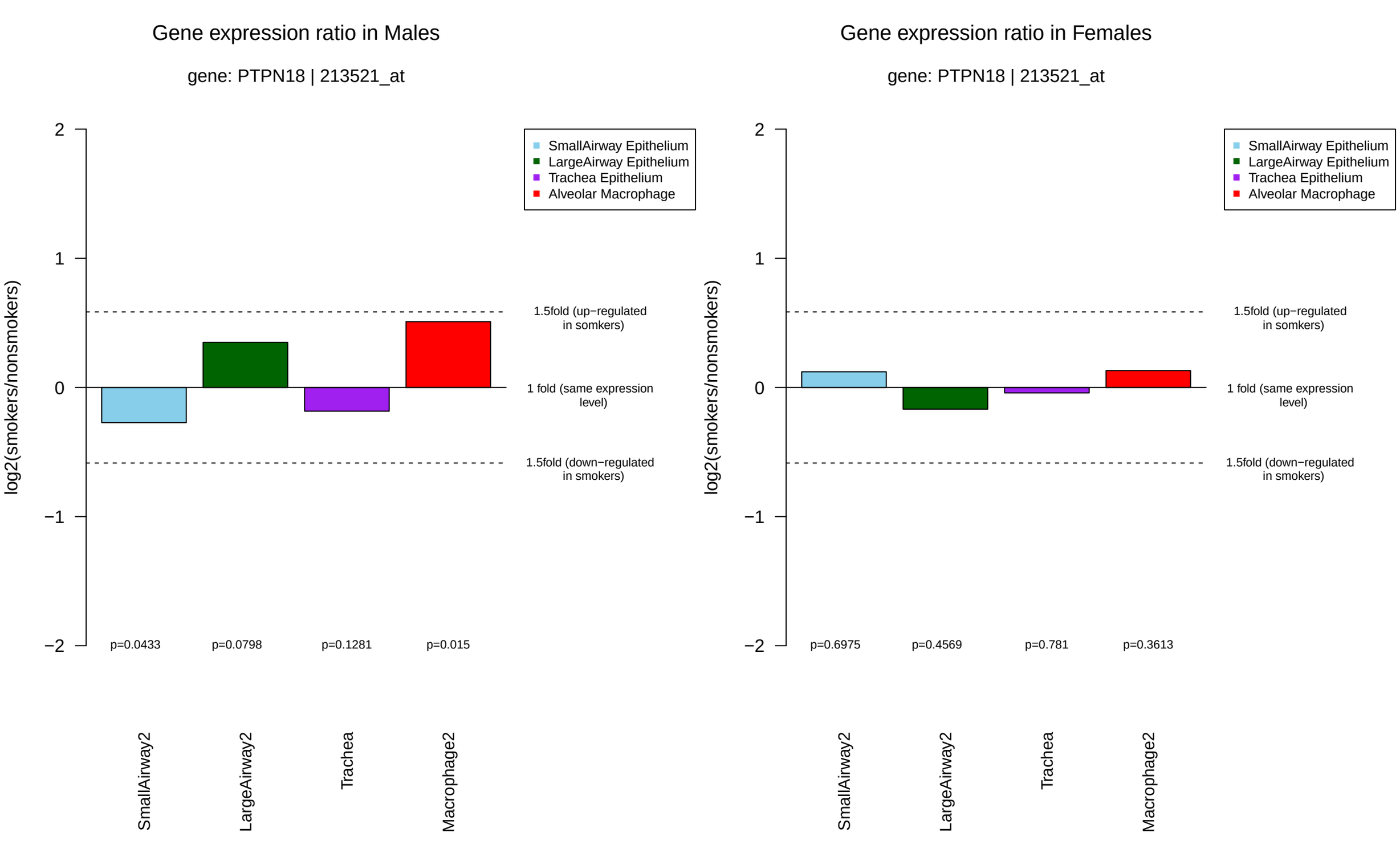
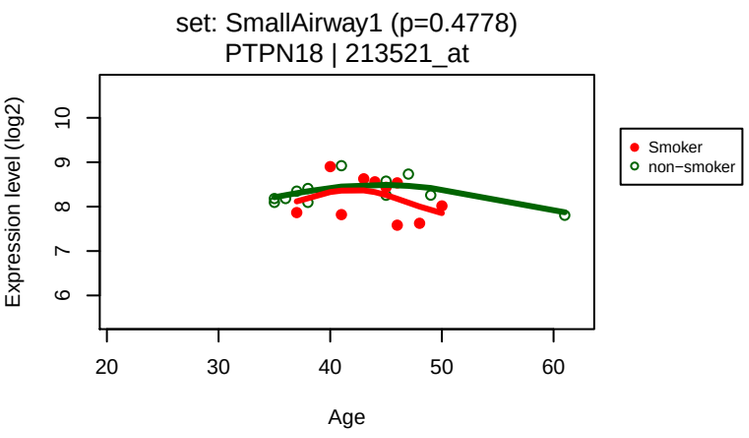
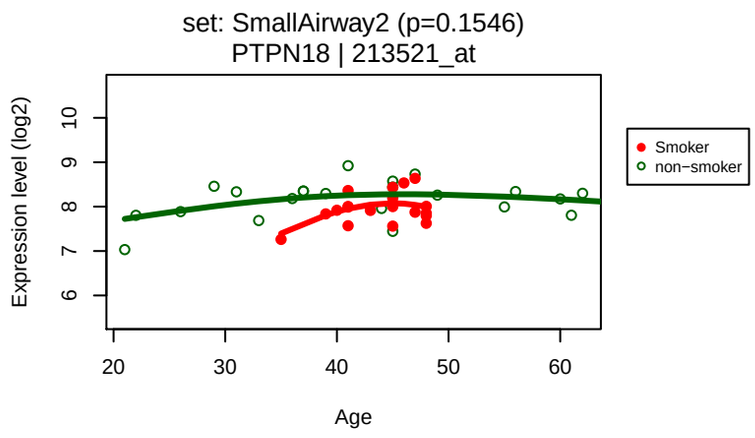
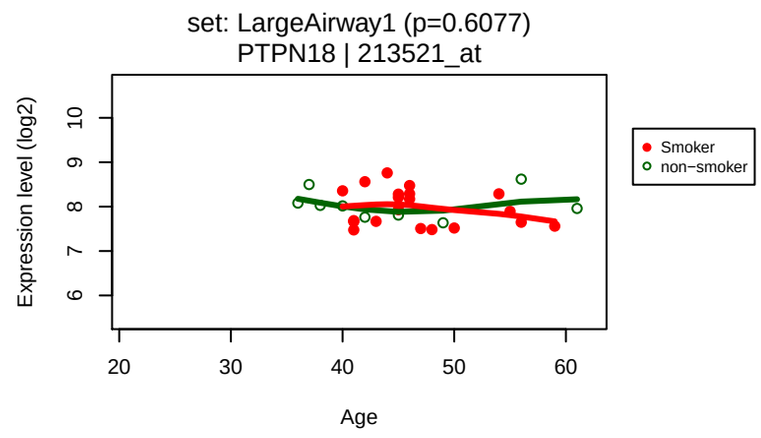

Gene expression of smokers and nonsmokers
gene: PTPN18 | 213521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4778
p=0.1546
p=0.4346
p=0.6077
p=0.2844
p=0.0883
p=0.6001
p=0.0091
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

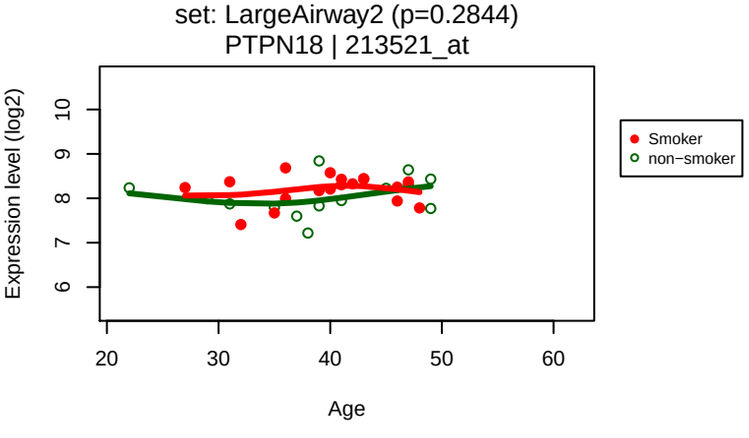

Gene expression ratio (smokers/nonsmokers)
gene: PTPN18 | 213521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4778
p=0.1546
p=0.4346
p=0.6077
p=0.2844
p=0.0883
p=0.6001
p=0.0091
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

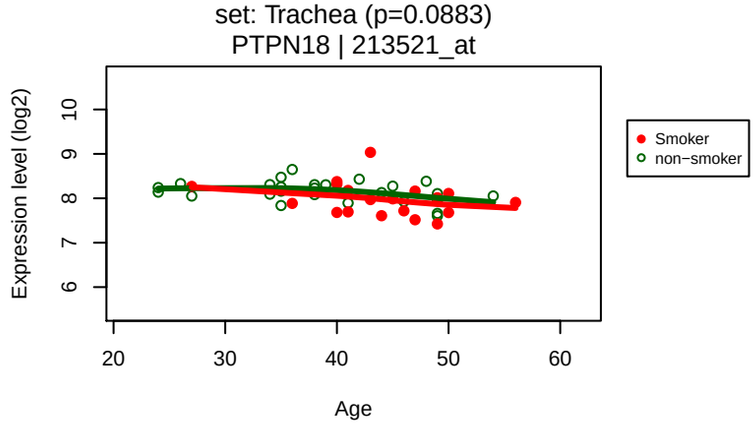

Gene expression ratio in Males
gene: PTPN18 | 213521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0433
p=0.0798
p=0.1281
p=0.015
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN18 | 213521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6975
p=0.4569
p=0.781
p=0.3613
SmallAirway2
LargeAirway2
Trachea
Macrophage2

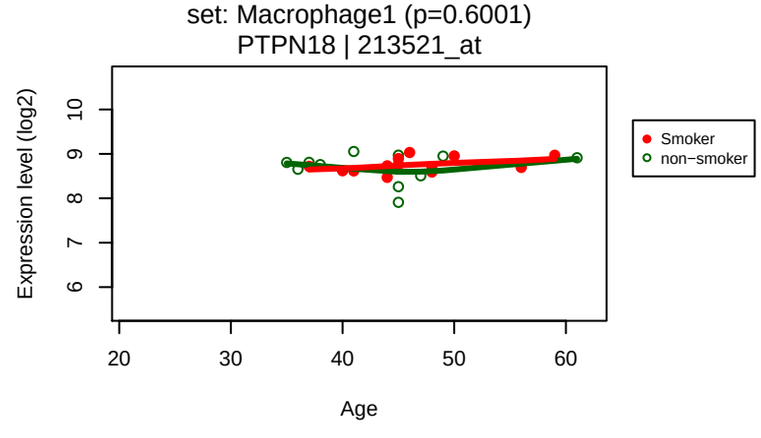

set: SmallAirway1 (p=0.4778)
PTPN18 | 213521_at
Expression level (log2)
Age
Smoker
non−smoker

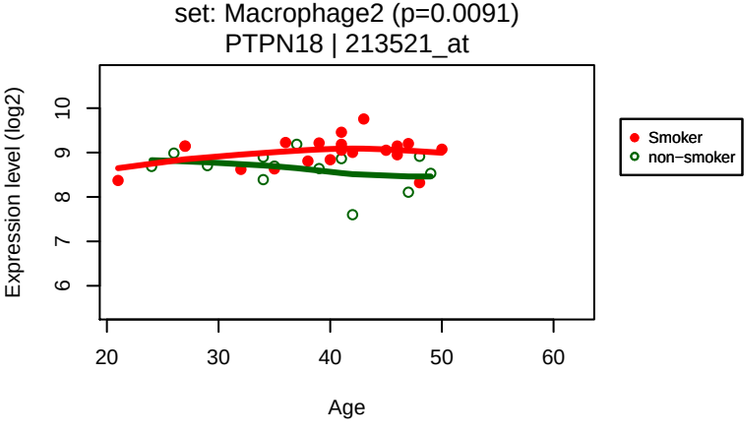

set: SmallAirway2 (p=0.1546)
PTPN18 | 213521_at
Expression level (log2)
Age
Smoker
non−smoker

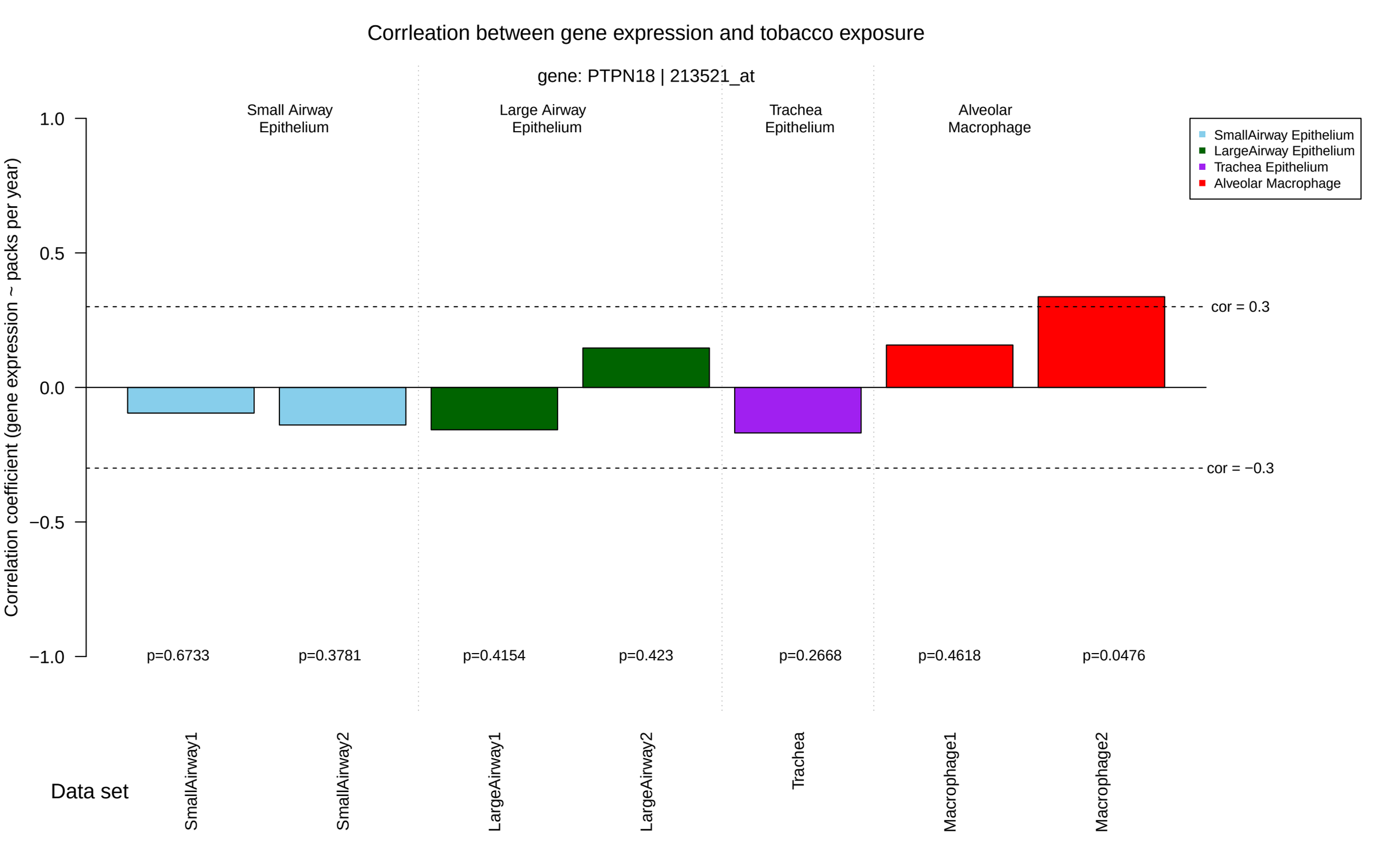

set: LargeAirway1 (p=0.6077)
PTPN18 | 213521_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2844)
PTPN18 | 213521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0883)
PTPN18 | 213521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6001)
PTPN18 | 213521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0091)
PTPN18 | 213521_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN18 | 213521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6733
p=0.3781
p=0.4154
p=0.423
p=0.2668
p=0.4618
p=0.0476
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2